

Exhaustive Capabilities List

Date: 2026-05-30
Organizations: alex-place, human-flourishing-frameworks
Scope: Complete inventory of all capabilities across all repositories
Status: Comprehensive, exhaustive listing

CORE CONTROL PLANE CAPABILITIES

Lantern OS (alex-place/lantern-os)

- ##### AI Control Capabilities
 - Local-first AI control plane
 - ASI Arc Reactor MK1 integration
 - Brier-style error tracking for confidence calibration
 - Human trial demo readiness gates
 - Evidence-based confidence scoring
 - Claim boundary enforcement (no unverified ASI capability claims)
 - Architecture reference integration (ASI patterns as design guidance only)
 - Distributed fleet governance
 - Agent network supervision
 - Safety automation gates
- ##### Safety Capabilities
 - Windsurf hooks system (pre/post action validation)
 - Pre-run command validation (demo safety)
 - Pre-MCP tool use validation (dangerous tools)
 - Pre-write code validation (system paths)
 - Pre-read code validation (sensitive files)
 - Pre-user prompt validation (unsafe patterns)
 - Post-action audit logging
 - Central audit trail
 - Human approval gates
 - Break-glass recovery paths
 - MCP canary validation
 - Safety gate enforcement
- ##### Convergence Capabilities
 - 12-step convergence loop
 - Actionable vs held issue classification
 - Fix window (4 issues prioritization)
 - Evidence receipts
 - Promotion states (theoretical → prototype → proven → production)
 - Rollback path documentation
 - Operator approval workflow
 - Dependency profiling
 - Adapter/manifest system
 - Promotion boundary enforcement
- ##### Documentation Capabilities
 - Markdown-based documentation system
 - PDF generation (perfect art style, simple style)
 - Report generation (convergence, capabilities, repo scan)
 - Evidence manifest tracking

- Validation receipt generation
- Audit log generation
- Skills documentation framework
- Workflow documentation

EXECUTION DEPENDENCY CAPABILITIES

gm-agent-orchestrator

Agent Orchestration

- Multi-agent coordination
- MCP (Model Context Protocol) connectors
- Agent fleet management
- Task pooling and batching
- Parallel processing with coordination
- Fallback and recovery paths
- Agent task distribution
- Agent state management
- Agent communication protocols

MCP Integration

- MCP connector framework
- Tool exposure validation
- MCP canary validation
- Tool safety checks
- Tool audit logging
- Tool permission management
- Tool fallback paths

gamemaker-room-editor

GameMaker Tooling

- Room editor tooling
- GameMaker integration
- Cloud-access for users
- Collaborative game development
- Asset management
- Room configuration
- GameMaker project management

lantern-symbolic-sandbox

Symbolic Reasoning

- Symbolic reasoning engine
- RAG quarantine system
- Evidence validation
- Content classification
- Symbolic logic processing
- Reasoning trace logging
- Symbolic decision support

Content Management

- Content intake system
- Rights review gates

- Evidence-based content promotion
- Quarantine for unverified content
- Content metadata management
- Content classification system
- Rights validation workflow

RAG DEPENDENCY CAPABILITIES

human-flourishing-frameworks/human-flourishing-frameworks

COMET LEAP Framework

- 30-day decision model
- Founder wisdom protocols
- Bayesian confidence updating
- Evidence-based decision loops
- Founder decision card extraction
- 30-day sprint methodology
- Confidence calibration
- Wisdom preservation
- Rapid iteration framework

Bayesian World Modeling

- Bayesian confidence tracking
- World model updates
- Evidence-based belief revision
- Confidence calibration protocols
- Bayesian decision support
- Probability estimation
- Uncertainty quantification

Human Flourishing Frameworks

- Flourishing metrics
- Evidence-based interventions
- Founder wisdom extraction
- Flourishing measurement
- Intervention effectiveness tracking
- Flourishing goal setting
- Progress measurement

BUSINESS SERVICE CAPABILITIES

moneybags

Wallet/Store Functionality

- Wallet management
- Store functionality
- Payment processing reference
- Transaction tracking
- Wallet/store design patterns

KeepSafe

Desktop Utility

- Desktop security utility
- File protection

- Desktop automation
- Security monitoring

LoL-Chat

Chat API Client

- League of Legends chat integration
- API client functionality
- Chat message handling
- API currentness management

Cloudbot

Cloud Bot Application

- Cloud bot framework
- Bot token management
- Environment configuration
- Cloud service integration

clouddataissexy

Cloud Data Application

- Cloud data processing
- Data application framework
- Environment configuration
- Cloud service integration

corporate_crawler

Web Crawling

- Corporate web crawling
- Endpoint discovery
- Rights review workflow
- Data extraction

OpenTray

Desktop Utility

- Desktop tray utility
- System tray integration
- Desktop automation

badslothserver

Server Application

- BadSloth server
- Secrets/environment management
- Server configuration

server-debugclient

Debug Client

- Server debug functionality
- Debug-only operations
- Debug logging

slothapi

API Server

- Sloth API server

- API endpoint management
- Environment configuration
- API validation

intspeedcheck

- ##### Utility
- Internet speed checking
 - Network performance monitoring
 - Utility reference

statusmonitor

- ##### Monitoring
- Status monitoring application
 - System health tracking
 - Monitoring fallback

smartmealplanning

- ##### Productivity
- Smart meal planning
 - Family productivity
 - Meal scheduling

SmartBid

- ##### SMB Tool
- Smart bidding application
 - SMB automation
 - Bidding optimization

place_co

- ##### Company Website
- Company public surface
 - Website management
 - Public content management

PAYMENT BRIDGE CAPABILITIES

Payment Bridge (apps/lantern-garage/payment-bridge)

- ##### Payment Processing
- Stripe integration
 - Payment processing
 - Invoice generation
 - Payment tracking
 - Transaction logging
 - Payment validation
- ##### Service Automation
- US Bank integration
 - Statement parsing
 - Cost transparency server
 - Service automation
 - Business automation
 - Invoice conversion

• Cost tracking
Financial Transparency
• Cost transparency server • Financial visibility • Cost tracking dashboard • Invoice management • Cash flow optimization

GAME DEVELOPMENT CAPABILITIES

GameMaker Projects

ChildOfLevistus
• GameMaker game project • Cloud-access for users • Game development tools • Asset management
GDXJam, GdxJam2
• GameMaker jam games • Game development • Asset preservation • Game project management
Other Game Projects
• Returners, Quest, Porkopolis, MysticGarden, BadSloth, Mr.Nom, GDungeon, Dungeon, Orion • Game development • Asset management • Game project preservation • Source-summary preservation

libGDX

Game Framework
• libGDX framework fork • Game development framework • Vendor-sized library management • Framework delta analysis

LIBRARY/VENDOR CAPABILITIES

riot-api-java, jriot

API Clients
• Riot Games API Java client • API client management • API currentness review • External API integration

The_Josephus_Problem

Algorithm Implementation
• Josephus problem implementation • Algorithm reference • Sample code

SITE CONTENT CAPABILITIES

alex-place.github.io

Personal Website

- Personal GitHub Pages site
- Public site reference
- Website management

TheGiddyLimit.github.io

Large Content Site

- Large GitHub Pages site
- Content management
- Rights review workflow
- Metadata management

dnd_homebrew

D&D Content

- D&D homebrew content
- Content rights review
- Content management

PLACEHOLDER CAPABILITIES

VisionPark, alex-place..github.io, tradiest, human-flourishing-frameworks/.github

Control Points

- Empty control points
- Organization configuration
- Profile management
- Placeholder for future use

RETIRED REFERENCE CAPABILITIES

HFF Recovery Snapshots

Evidence Preservation

- hff-lantern-recovery (recovery evidence)
- hff-evidence-master-clean (evidence master)
- hff-master-clean (master clean evidence)
- hff-release-candidate (release candidate evidence)
- hff-seven-validate (validation comparison)
- Evidence-only reference
- Recovery documentation
- Comparison snapshots

Local Backups

Backup Evidence

- Local backup directories
- Recovery evidence
- Backup documentation

NOVEL RESEARCH CAPABILITIES

ASI Pattern Integration

- Architecture references only
- Strict claim boundaries
- Evidence-based promotion
- Brier-style error tracking
- Human trial readiness gates
- Safe ASI pattern adoption
- Regulatory compliance preparation

Bayesian Founder Wisdom

- Founder wisdom as structured data
- Bayesian confidence updating
- Evidence-based promotion
- 30-day sprint methodology
- Decision card extraction
- Wisdom preservation
- Wisdom democratization

Safety-First Agent Orchestration

- Pre/post hooks for all actions
- MCP canary validation
- Audit logging
- Human approval gates
- Break-glass recovery
- Safe agent operations
- Transparency
- Human control

Evidence-Based Convergence

- 12-step convergence loop
- Actionable vs held issues
- Fix window
- Evidence receipts
- Promotion states
- Evidence-based decisions
- Risk mitigation
- Rollback paths

Human Flourishing Metrics

- Flourishing metrics
- Evidence-based interventions
- Founder wisdom protocols
- Bayesian world modeling
- COMET LEAP methodology
- Flourishing measurement
- Intervention effectiveness

INTEGRATION CAPABILITIES

Lantern OS + COMET LEAP Integration

- Brier-style error tracking (from COMET LEAP)
- Founder wisdom protocols (from COMET LEAP)
- 30-day sprint methodology (from COMET LEAP)

- Human trial readiness gates (from Lantern OS)
- Safety automation (from Lantern OS)
- Evidence-based AI control with safety gates

Agent Orchestration + Symbolic Sandbox

- MCP connectors provide tool access
- Symbolic sandbox provides content validation
- Safe tool use with content validation
- Audit logging for transparency
- Human approval gates
- Safe agent operations with content validation

Business Services + Payment Bridge

- Business services provide automation
- Payment bridge provides transaction processing
- Cost transparency server provides visibility
- Automated business operations with financial transparency
- Financial audit trails

Game Development + Cloud Access

- Game development tools
- Cloud-access for users
- Collaborative game creation
- Asset preservation
- Community game projects

Safety Hooks + Audit Logging

- Comprehensive hook coverage
- PowerShell validation scripts
- Audit logging to central log
- Exit code 2 for blocking
- Human trial readiness integration

THEORETICAL APPLICATION CAPABILITIES

Personal AI Assistant

- Local-first processing (no cloud dependency)
- Safety hooks for dangerous operations
- Human approval for sensitive tasks
- Audit logging for transparency
- Evidence-based confidence
- Privacy by design
- Safety by design
- Transparency by design

Family Coordination System

- Family task coordination
- Decision support with evidence
- Budget tracking and transparency
- Communication logging
- Conflict resolution support
- Family-specific workflows
- Evidence-based decisions

- Financial transparency

Small Business Automation

- Payment processing
- Service automation
- Cost tracking
- Invoice generation
- Safety gates for financial operations
- Cost transparency
- Invoice automation
- Financial audit trails

Educational Tools

- Learning content delivery
- Progress tracking
- Capability honesty (no overclaiming)
- Evidence-based assessment
- Safety for student use
- No overclaiming
- Progress tracking

Creative Work Augmentation

- Creative tool integration
- AI assistance for creative tasks
- Human-in-the-loop control
- Attribution and credit
- Safety for creative workflows
- Augmentation not replacement
- Human-in-the-loop
- Attribution systems

CAPABILITY MATURITY

Theoretical (All Capabilities)

- All capabilities documented
- Theoretical frameworks defined
- Research directions identified
- Novel approaches documented

Prototype (Most Capabilities)

- ASI Arc Reactor MK1
- Agent orchestration
- Business services
- Symbolic sandbox
- COMET LEAP framework
- Human flourishing frameworks
- Safety hooks
- Payment bridge

Proven (Core Capabilities)

- Game development tools
- Convergence loop
- Safety hooks

- Cloud-access for games
- PDF generation
- Documentation system

Production (Limited)

- Game development
- Convergence loop
- PDF generation
- Documentation system

COMPLETE CAPABILITY COUNT

Category	Count
Core Control Plane	30+
Execution Dependencies	20+
RAG Dependencies	15+
Business Services	40+
Game Development	30+
Library/Vendor	10+
Site Content	10+
Placeholder	5+
Retired References	10+
Novel Research	10+
Integration Capabilities	10+
Theoretical Applications	20+
Total	210+

Report Generated: 2026-05-30
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Scanner: Cascade (Lantern OS Agent)
Status: Exhaustive capability listing complete